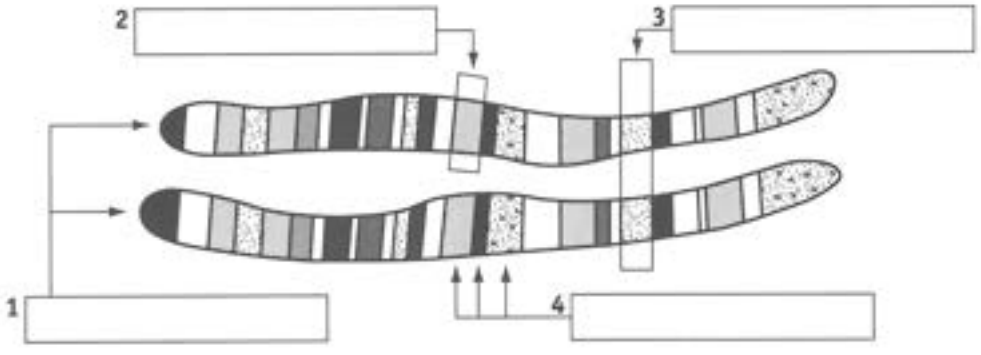
Label the parts (1 to 4) on this diagram using:

- allele
- allele pair
- gene locus
- homologous pair



Match the terms used in the chromosome diagram above with their correct definitions below:

5 One half of a complete set is derived from each parent. _____

6 Pair of genes required to supply the code for a specific trait. _____

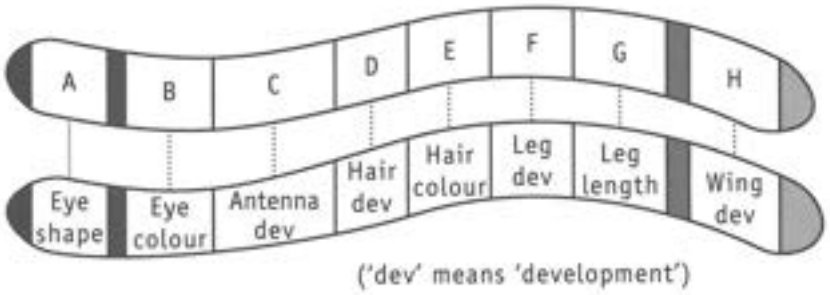
7 The exact location of a gene on a chromosome. _____

8 One of several forms of a gene for a characteristic. _____

Mapping

In the same way that cartographers identify and draw land masses when map-making, genetic maps can also be designed. As researchers discover which genes on which chromosomes are responsible for specific traits, the loci (plural of *locus*) are given numbers and a map is created. When homologous pairs align correctly, coding for a particular character occurs.

Study the hypothetical homologous pair below. The letters correspond to the traits described on the same locations on the other half of the pair.



('dev' means 'development')

9 Which locus on the diagram above is responsible for leg length? _____

10 Which locus is responsible for eye shape? _____

11 Which locus is responsible for antenna development? _____

12 Which locus is responsible for leg development? _____

13 Which trait would be affected if the alleles at locus B were not expressed? _____

14 Which trait would be affected if the alleles at locus E were not expressed? _____

15 Which trait would be affected if the alleles at locus D were not expressed? _____

16 Which trait would be affected if the alleles at locus H were not expressed? _____

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